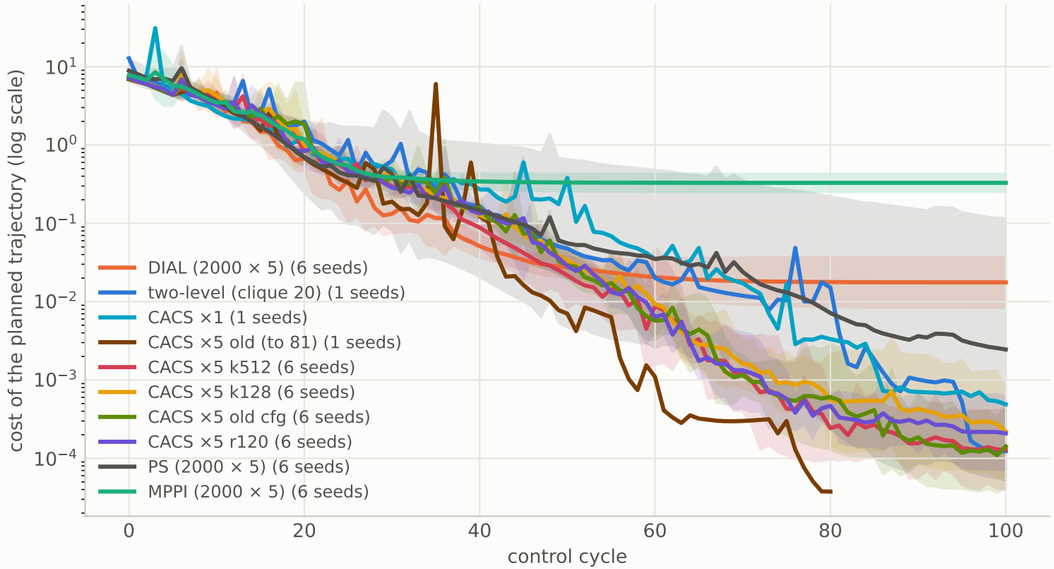
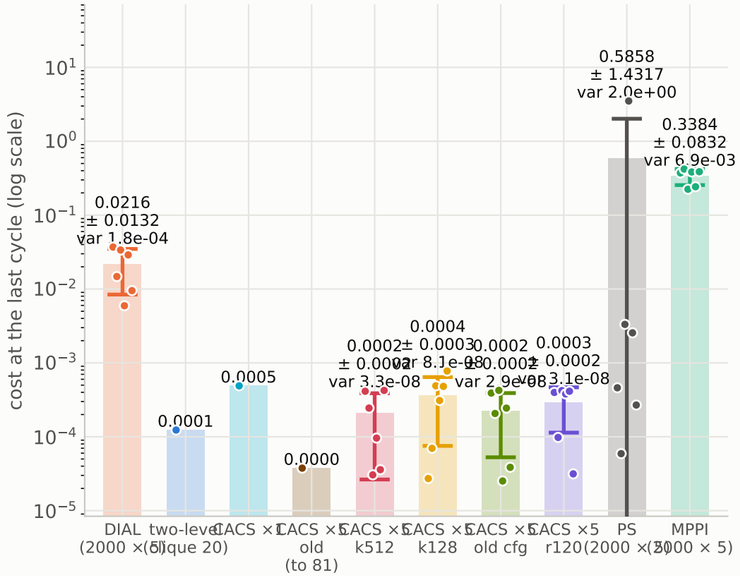
PushT closed loop: GlobalMPPI (KSOS ADMM-cuda + MPPI σ 0.1) vs PS / MPPI / DIAL

start: block (0.1, 0.1, 1.3), pusher (0, 0); 100 cycles at 50 Hz. GlobalMPPI variants (seed 0 only, except CACS ×5 k512 / k128 / old cfg / r120: seeds 0–5), each KSOS on ADMM-cuda then MPPI 1000 × 5, σ 0.1: two-level Sobol, 5 restarts × 1000 samples in shrinking boxes, cliques of 20; CACS ×1, one cost-aware chordal sampling pass over the whole box per cycle (≈ 1,200 rollouts); CACS ×5 + warm, one CACS pass per restart in the shrinking boxes, started from the previous plan, old config h 1.2, k 128, 40 replacements (crashed at cycle 81) and new config h 0.7, reuse 40, batch 768, no replacements, k 512 or k 128 (100 cycles, seeds 0–5); old cfg = old hyperparameters h 1.2, k 128, 40 replacements on the current code; r120 = k 512 with reuse 120 (seeds 0–5). Final cost = cost at the last cycle reached. PS / MPPI / DIAL: pusht.py settings with 2000 × 5 (≈ 10,000 rollouts), seeds 0–5. Time statistics leave out cycle 0 (JIT); std uses ddof = 1

Cost curve: geometric mean ± 1 std (log) over seeds



Final cost: mean ± std over seeds



Planning time per cycle: mean ± std over seeds

